Curvature and Torsion in the Continuum Theory of Lattice Defects

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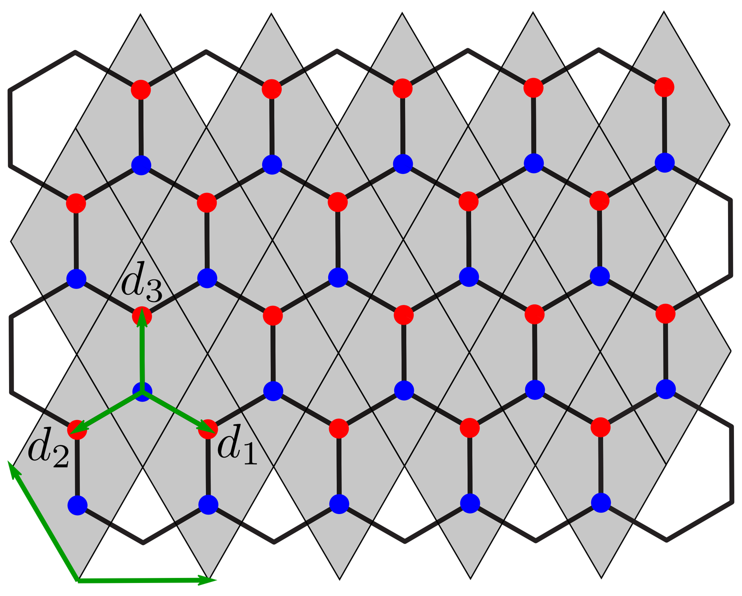
Offen im Denken

Continuum Theory - Graphene [1]



Tight-binding Hamiltonian

$$H = \sum_{\vec{r}} \sum_{j=1}^{3} \left(a^\dagger_{\vec{r}} b_{\vec{r}+\vec{d}_j} + b^\dagger_{\vec{r}+\vec{d}_j} a_{\vec{r}}\right)$$

Fourier transform

$$i\partial_t \vec{\psi}(\vec{k}) = \boldsymbol{h}(\vec{k})\vec{\psi}(\vec{k})$$

$$\boldsymbol{h}(\vec{k}) = \begin{pmatrix} 0 & f(\vec{k}) \\ f^*(\vec{k}) & 0 \end{pmatrix}, \quad f(\vec{k}) = \sum_{j=1}^{3} e^{i\vec{k}\cdot\vec{d}_j}$$

Linear dispersion around Dirac points

$$\vec{k} = \vec{K}^{(s)} + \vec{\delta k} \quad \Longrightarrow \quad \boldsymbol{h}(\vec{k}) \approx -\frac{3}{2}\vec{\delta k}\cdot\vec{\boldsymbol{\sigma}}$$

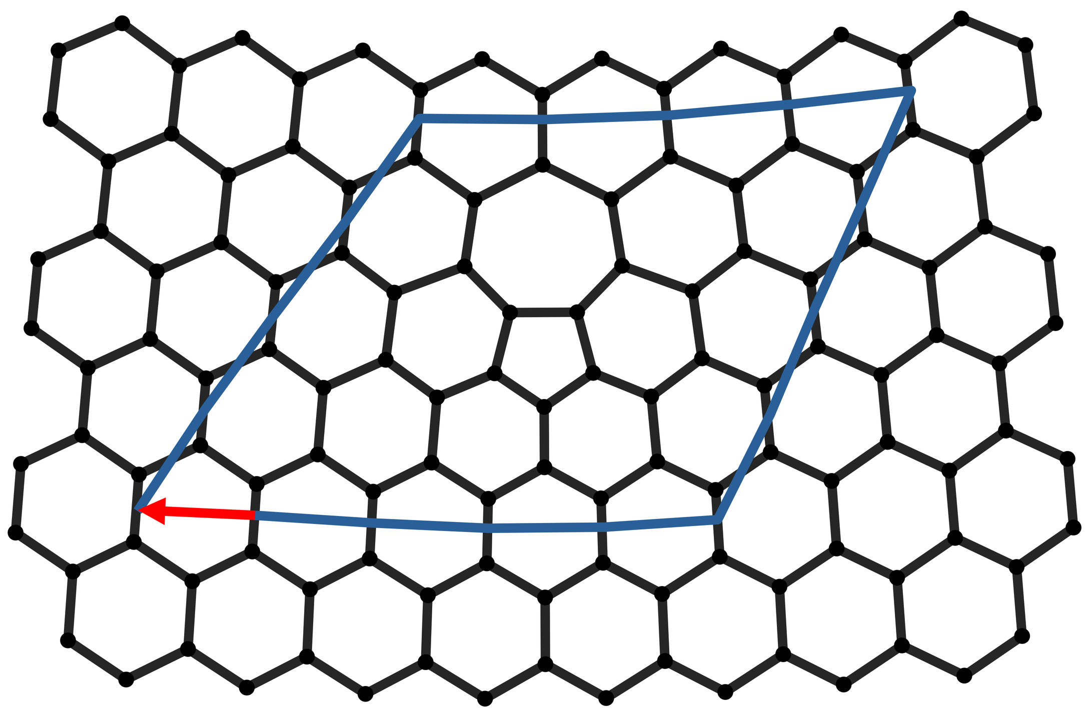
Continuum limit

$$i\partial_t\vec{\phi}(\vec{r}) = i\,c\,\boldsymbol{\sigma}^i\partial_i\vec{\phi}(\vec{r}), \qquad \rho = \vec{\phi}^\dagger\vec{\phi}, \qquad \vec{j} = -c\vec{\phi}^\dagger\vec{\boldsymbol{\sigma}}\vec{\phi}$$

$$\Updownarrow$$

$$i\gamma^\mu\partial_\mu\phi = 0, \quad \gamma^\mu = \sigma^\mu\sigma^3, \quad \mu \in \{0,1,2\}$$

Burgers vector in 5-7 defects [2]



One 5-7 defect causes a lattice dislocation which can be described by the Burgers vector $\vec{b} = -\sqrt{3}\,\vec{u}_x$.

Torsion [3]

In standard GR the torsion is assumed to vanish. It is defined as the antisymmetric part of the connection:

$$T^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\mu\nu} - \Gamma^\lambda{}_{\nu\mu}$$

Any general affine connection can be decomposed into a symmetric part, the contorsion and the disformation:

$$\Gamma^\alpha{}_{\mu\nu} = \{^\alpha{}_{\mu\nu}\} + K^\alpha{}_{\mu\nu} + L^\alpha{}_{\mu\nu}$$

Here we assume **metric-compatibility**:

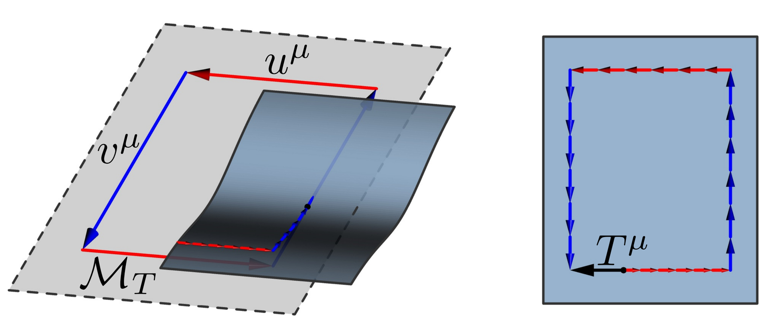
$$\nabla_\lambda g_{\mu\nu} = 0 \Leftrightarrow L^\alpha{}_{\mu\nu} = 0$$

The **contorsion tensor** is build from the torsion:

$$K^\alpha{}_{\mu\nu} = \frac{1}{2}\left(T^\alpha{}_{\mu\nu} - T_{\mu\nu}{}^\alpha + T_\nu{}^\alpha{}_\mu\right)$$

Torsion can be thought of as the displacement vector $T^\mu(u,v)$ resulting from rolling the tangent plane along a parallelogram:

$T^\mu(u,v)$
$= u^\lambda v^\nu T^\mu{}_{\nu\lambda}$

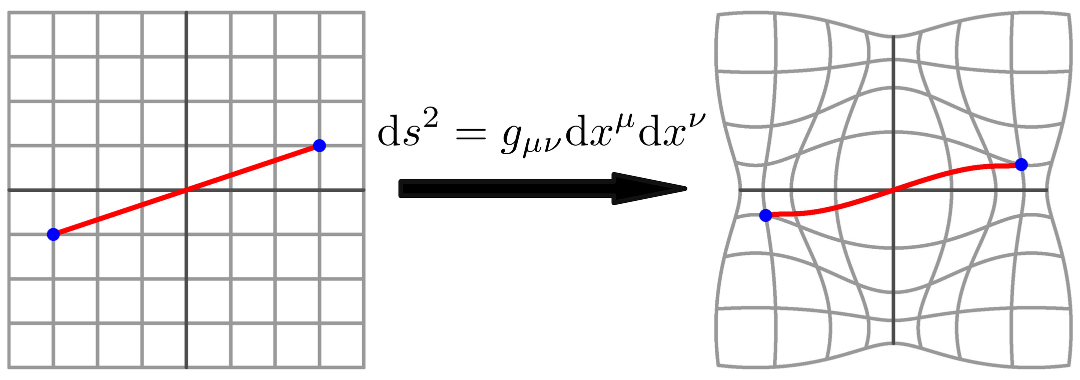


Riemannian Geometry [3]

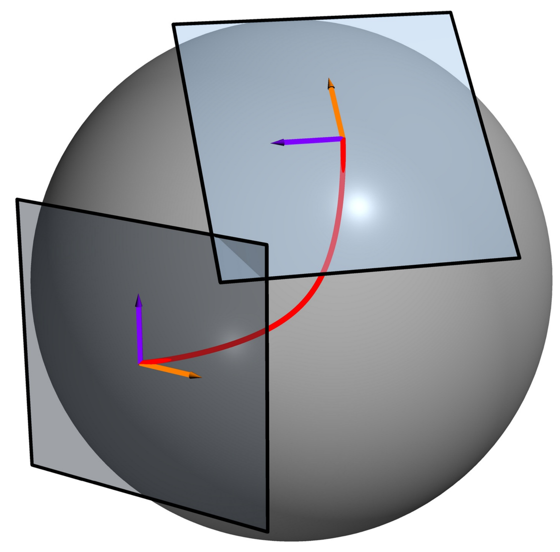
In GR we use a Riemannian manifold $\mathcal{M}$ equipped with metric $g_{\mu\nu}$ and connection $\Gamma^\lambda{}_{\mu\nu}$.

Metric

Defines distances and angles on the manifold.



Connection



Connectes nearby **tangent spaces**, which allows the definition of a derivative on **vector fields** which take on values in the tangent space.

The unique torsion-free, metric compatible connection is called the **Levi-Civita connection**:

$$\{^\lambda_{\mu\nu}\} = \frac{1}{2}g^{\lambda\rho}\left(\partial_\nu g_{\rho\mu} + \partial_\mu g_{\rho\nu} - \partial_\rho g_{\mu\nu}\right)$$

Covariant derivative

Takes into account the structure of the manifold and transforms like a tensor.

$$\nabla_\mu v^\nu = \partial_\mu v^\nu + \Gamma^\nu{}_{\mu\lambda}v^\lambda$$

Autoparallels

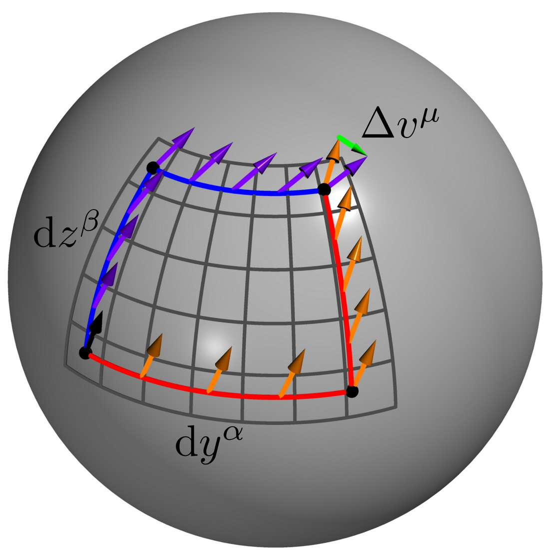
A curve whose tangent vector is parallel-transported along itself.

$$u^\mu\nabla_\mu u^\alpha = 0$$

Curvature

The Riemann curvature tensor describes how parallel transport of vectors depends on the path.

$$\Delta v^\mu = R_{\alpha\beta}{}^\mu{}_\nu\, v^\nu\, dy^\alpha\, dz^\beta$$



It can be expressed in terms of the connection:

$$R_{\alpha\beta}{}^\mu{}_\nu = \partial_\alpha\Gamma^\mu{}_{\beta\nu} - \partial_\beta\Gamma^\mu{}_{\alpha\nu} + \Gamma^\mu{}_{\alpha\lambda}\Gamma^\lambda{}_{\beta\nu} - \Gamma^\mu{}_{\beta\lambda}\Gamma^\lambda{}_{\alpha\nu}$$

Metric $\leftrightarrow$ torsion mapping in 2D

In 2D we propose a physical mapping between a **classical** Manifold $\mathcal{M}_g(g_{\mu\nu}, \{^\lambda_{\mu\nu}\})$ and **purely torsionful** Manifold $\mathcal{M}_T(\delta_{ij}, K^k{}_{ij}\})$.
In 2D we can always choose coordinates in which the metric is $g_{\mu\nu} = \delta_{\mu\nu}\,e^{2\phi(x,y)}$.
Also the torsion in 2D can be written using a single covector $T^k{}_{ij} = b^k\varepsilon_{ij}$
We demand equal **curvature** of both manifolds:

$$R\left[\mathcal{M}_g\right] = R\left[\mathcal{M}_T\right],$$

This leads to the following solution for the torsion vector b^i as a function of the metric ϕ and an arbitrary scalar field χ:

$$b^i = \varepsilon_{ik}\,\partial_k\phi + \partial_i\chi$$

For $\chi = 0$ the **autoparallels** of both manifolds are equivalent up to a reparameterization. It can be shown, that the χ field corresponds to local roatations of the frame field.

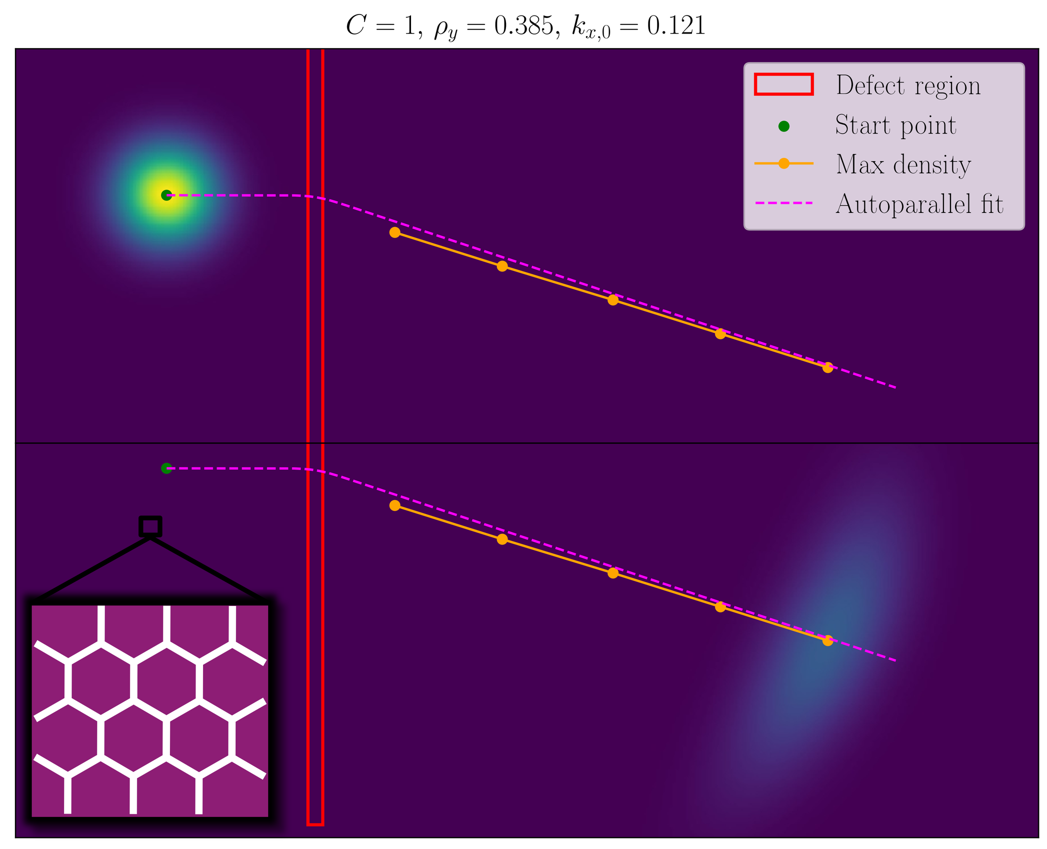
Conclusion

Except for distances and the problem of different tangent spaces a physical mapping between torsion and metric is possible.

Tight-binding numerics

Using a **vector representation** containing the wavefunction at the discrete lattice sites, the Hamiltonian is a **sparse Matrix** acting on this vector to generate the time evolution:

$$i\partial_t\vec{\psi} = H\vec{\psi} \quad \Rightarrow \quad \vec{\psi}(t) = e^{-iHt}\vec{\psi}_0$$



Torsion model

Take $\mathcal{M}_T$ with $b^1(x)$ and $b^2 \equiv 0$. The autoparallel equation becomes:

$$\ddot{x} = -\dot{x}\dot{y}b^1, \quad \ddot{y} = \dot{x}^2 b^1$$

Since $\frac{d}{dt}(\dot{x}^2 + \dot{y}^2) = 0$ we can substitute:

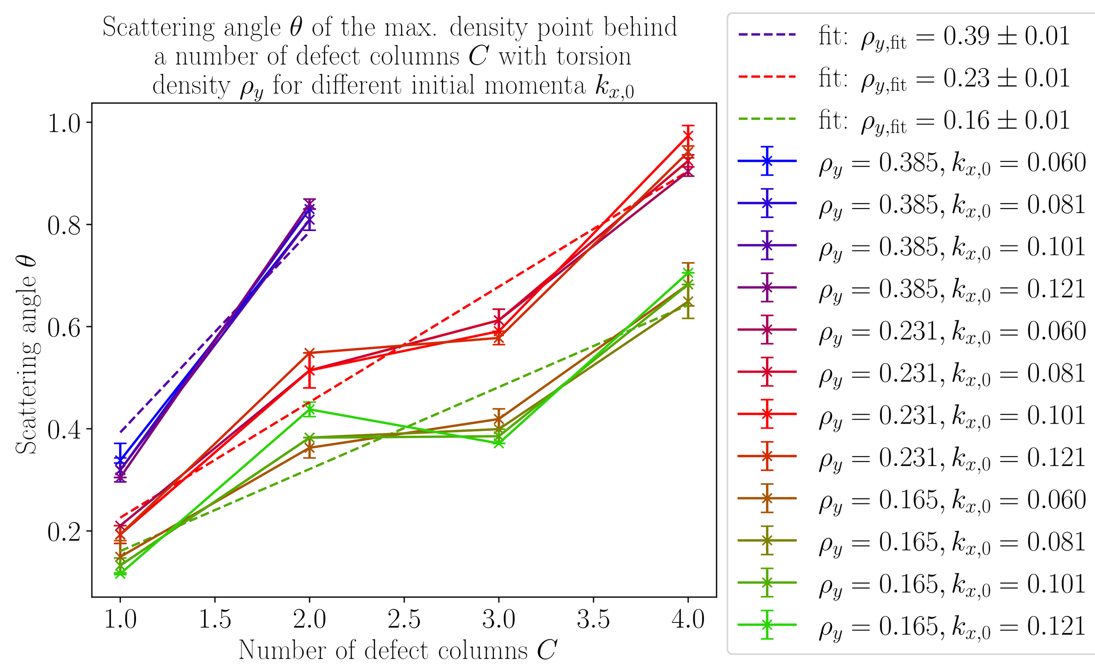
$$\dot{x} = \cos(\theta), \qquad \dot{y} = \sin(\theta)$$

The autoparallel equation becomes $\dot{\theta} = b^1(x)\cos(\theta)$. Eliminating the time dependence yields $\frac{d\theta}{dx} = b^1(x)$. We take $\dot{x}(x=0) = 1$, $\dot{y}(x=0) = 0$. With this we can calculate the scattering angle of the torsion model:

$$\theta_T = \int_0^\infty b^1(x)\,dx$$

Interpreting torsion as a Burgers vector density with distance Δy between defects gives:

$$\rho_y = (\vec{b})_x/\Delta y \approx \frac{\iint b^1 dxdy}{\Delta y} = \int b^1 dx = \theta_T$$



Outlook

The classical Dirac equation obtained from continuum theory of graphene can be generalized to describe **curved graphene** by simple generalization to a non-flat geometry [1]. The problem is, that this mechanism only works for a torsionless geometry. For a geometry with torsion, which is used to model the scattering with autoparallels, the Lagrangian needs to be constructed in a way that preserves **hermiticity** [4]. The question is whether the dynamics resulting from this Lagrangian can describe the tight-binding results.

References

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[4] de Juan F, Cortijo A, Vozmediano MAH. Dislocations and torsion in graphene. *Nucl. Phys. B* **828**, 625 (2010).